

Passive Optical Networks (PONs): Past, present, and future

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ABSTRACT

Optical access solutions have attracted the attention of researchers from both academia and industry for a long time. In the past these solutions were not cost effective for service-provider deployment. This situation has been changing over recent years. Passive Optical Networks (PONs) represent one of the most attractive optical access-network solutions. In this paper, we examine the history of PONs, investigate their current status, and explore their future opportunities. The focus of our review is on PON standards and on deployment trends.

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1. Introduction

The tremendous growth of the Internet, coupled with new generation of services and applications, continue to change the landscape of telecommunications worldwide. Access networks have been a special focus of this change over recent years. An access network is referred to as a last-mile (or sometime first-mile) network since it comprises the segment of a service provider's network which connects end users (customers) to nearest provider's node/equipment, be it a Central Office (CO) or a remote feeder node. Although research in fiber-based access networks dates back to more than twenty years, optical access solutions have not been seriously considered for deployment in service-provider networks until few years ago. Indeed, copper (twisted-pair and coaxial cables) still represent the dominant transmission medium in today's

access networks, globally speaking. This is the case today in both telecom and cable-TV networks, but the scene could be changing in the near future.

With progress in digital telephony, data communications, and the emerging Internet; broadband access started to become desirable around mid 1980's. Several solutions were envisioned at that time, including fiber-based ones, but the economics of optical access were not encouraging. The Integrated Service Digital Network (ISDN) was probably the first standard for broadband access. The copper-based ISDN enjoyed some success only in few areas of the world. Cost was an issue for this technology and so was the lack of serious driver applications.

Widespread deployment of broadband access unfolded with the rising popularity of the Internet and the introduction of the Digital Subscriber Line (DSL) and Cable Modem standards. DSL uses twisted-pair copper cables and can have different flavors, which are referred to collectively as xDSL. While the lower frequency range of the copper pair bandwidth is reserved for the 4 KHz traditional telephony, its higher-frequency range is used for data transmission. Typically, the 25–160 KHz band is designated for

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upstream traffic (end user to service-provider network) and the 240 kHz to 1.5 MHz band is dedicated for downstream traffic (provider's network to end user). Another popular means of broadband access, especially in the US, is Cable Modems. Cable TV networks are primarily based on Hybrid Fiber Coaxial (HFC) architectures. Several video channels are multiplexed and used to modulate an optical source. An optical feeder line carries these broadcast channels from the provider's headend to remote hub nodes where signals are converted back to the radio frequency domain and transmitted over coaxial cables to end users. In the 1990's, US Cable companies, also known today as Cable Multiple System Operators (MSOs), started to change their one-way broadcast architectures into two-way communication networks by using bidirectional RF amplifiers and return optical links. With cable modems, these networks became able to deliver data services to end users. Although the maximum cable-modem bandwidth can reach 38 Mb/s downstream and 10 Mb/s upstream, most US Cable MSOs provide their customers a maximum of 3–8 Mb/s downstream and 0.2–0.8 Mb/s upstream [1].

There are a number of reasons why service providers have always been interested in optical access and in extending fiber reach towards customer premises. First, the information-carrying capacity of optical fiber is well beyond any predictable access demands. Today, several Gigabit-per-second upstream and downstream bandwidths are possible and more is envisioned in future. Thus, fiber is 'future-proof' from a bandwidth perspective. Due to its low loss, fiber is intrinsically long reach. It can deliver to end users without en-route electronics and power supply. This simplifies the outside distribution plant and eases operation and maintenance. Fiber is also immune to electrical interference, noise, and other external signal disturbances. It is more resistant than copper to environmental and chemical effects. As a matter of fact, a fiber plant is projected to last well over 25 years, and should have little trouble during that life. In the early days of optical access, the high cost of equipment (nearly US\$ 5000 per customer) was a problem. This is no longer the case.

A number of factors have actually started to change the economics of optical access since the late 1990's. Progress in optical technologies brought down component and system cost. In particular, areas such as optoelectronic and optical device fabrication, optical assembly and packaging, and ASICs (Application Specific Integrated Circuits) have seen considerable cost reductions. In the US, the 1996 Telecom Act opened up the field for Telcos and Cable MSOs to compete in the access domain and on each other's traditional territories. Demand for a new generation of bandwidth-hungry services and triple-play delivery to customers (voice, data, and video) ignited fierce competition among US service providers of all kinds. Competition among service providers of different types also started in many other countries by virtue of privatization and deregulation.

These developments motivated service providers worldwide to invest in FTTx (Fiber To The x). FTTx is an acronym that embraces a number of optical access technologies, such as Fiber To The Node (FTTN), Fiber To The Curb (FTTC),

Fiber To The Business – or Building – (FTTB), Fiber To The Home (FTTH), and Fiber To The Premises (FTTP). Depending on the physical point in the field where fiber terminates and interfaces with copper and/or equipment, FTTx may take any of these flavors and architectures. Different FTTx architectures can be realized using different protocols and technologies. Indeed, FTTx can be classified using a couple of other methods. Depending on the use of passive or active devices, an FTTx network is based on a Passive Optical Network (PON), or an Active Optical Network (AON). From a topological point of view, FTTx solutions can be categorized also into two types: Point-to-Point (P2P) systems and Point-to-Multi-Point (P2MP) systems [2].

In a P2P topology, a separate fiber runs from an Optical Line Terminal (OLT) in the Central Office (CO) to an Optical Network Unit (ONU) at some end point (node, curb, home, or otherwise). A remote Ethernet-based aggregation node can be placed between the OLT and the ONU to share the bandwidth of the feeder fiber cable among several end users. In this case, each user is still served by a P2P link (and has an ONU). The Ethernet node is electrically powered and utilizes active electronics. Therefore this solution is also said to be based on an AON. In architectures of this type, subscribers can be several tens of kilometers away from the remote node and several hundreds of customers can be supported [2,3]. In addition to major service providers, many small providers and municipalities deploy AON Ethernet based networks, where a small number of Ethernet switches are interconnected with fiber. This model is suitable for enterprise-type networks with little management overhead. Nevertheless, for these very reasons, such networks can be difficult to scale into some large-service-provider deployments.

In a P2MP topology, the feeder fiber runs from the OLT of the CO up to a splitting point in the field. From this point, shorter fibers extend to several ONUs at end points (node, curb, home, or otherwise). Passive optical splitters are used at the splitting point to distribute the input signal power among a number of outputs. The network utilizes no active electronics between the OLT and ONT units, so it is PON based. The nature of PONs, i.e. restricted in power-budget, makes this type of network relatively limited in range to 10–20 km. Tens of customers may be served by a PON [2,3]. Fig. 1 depicts a simple comparison between the AON and PON concepts.

FTTx fits well into the typical telecom access network topology of tree-and-branch, where the splitters/switches replace copper cross-connects and digital loop carriers. FTTx is also similar to a cable HFC distribution system. In contrast to FTTx, HFC systems are inefficient. They involve many legacy modulation formats and network elements. As the number of data-service users and their bandwidth consumption increases, HFC suffers from serious reduction in the spectrum set aside for video services.

Thus, progress in technology, new service offerings, demand for triple play, and policy changes have made a new case for FTTx. Service providers worldwide have made inroads towards large-scale deployment of FTTx. Subscriber numbers and take rates are up in the United States, DSL subscriber numbers have started to drop in China because of optical competition, and European

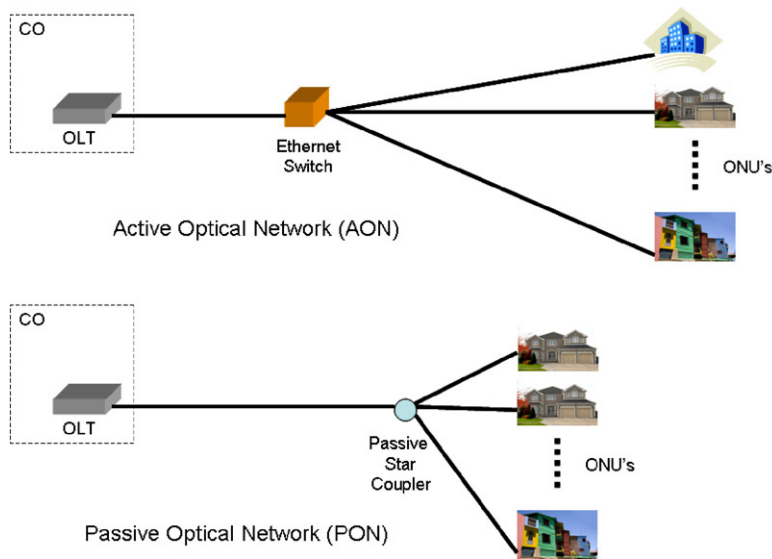


Fig. 1. Active and passive optical access network concepts.

incumbents and municipalities are moving ahead in FTTx deployment. Hence, a new generation of broadband optical access networks is starting to enable unprecedented family of novel services and applications [2].

In the remainder of this paper, we focus on PONs as an important FTTx architecture. We discuss progress in this field, PON standards, and deployment trends. PONs were probably the first FTTx technology. They have a number of uniquely attractive features. They require less equipment count and space, and consume less power than all competitive technologies. Passive statistical multiplexing enables PONs to offer higher peak bandwidth to end users. PONs utilize baseband digital signaling and do not require legacy modulation formats to deliver video services. One of the key issues in designing a PON is to devise a fair mechanism for users to share the feeder fiber bandwidth in the upstream direction. Three multiplexing schemes are possible: Time Division Multiplexing (TDM), Sub-Carrier Multiplexing (SCM), and Wavelength Division Multiplexing (WDM). To date, standard PON technologies are developed based on different forms of TDM. WDM PONs and TDM/WDM PONs are considered for future technology generations.

In the following section, we discuss the early days of PON deployment and introduce the concepts of Asynchronous Transfer Mode (ATM) PONs (A-PON). Then we examine the evolution of A-PON to Broadband PON (B-PON). This is followed by investigation of the current Ethernet PONs (E-PON) and Gigabit-capable PONs (G-PON). Then, we explore future trends and conclude the paper by a US perspective of some of the factors triggering future technologies and deployments.

2. The early PON trial period

PON was first introduced in the late 1980's [4–6]. At that time, optical technology was fairly new and beginning to be used in long-haul telecommunication networks.

Fiber optics enjoyed tremendous success in core and metro Telcos' networks and in cable backbones. By the mid 1990's, with fast growth in bandwidth demand, the case could be made for using optics in last-mile/access networks as well. This ushered in a period of standards development and field trials that lasted about 5 years.

One of the most important events in the development of PONs was the founding of the Full Service Access Network (FSAN) consortium in 1995 [7]. One of the goals of this ad-hoc organization of service providers, from around the world, was to create the economy of scale and lower the cost of optical access systems by promoting common standards. This included DSL and PON based solutions, and every combination thereof. The FSAN group was organized into separate working groups, with the Optical Access Network (OAN) working group being concerned with PON development. This group is the longest lived part of FSAN, having outlasted the Full Service Very high bit-rate Digital Line Subscriber (FS-VDSL), optical access management, and home networking groups. The OAN group has also been the most productive, having originated over ten independent numbered standards which are adopted by the International Telecommunication Union Standardization Sector (ITU-T).

At the beginning of the work on optical access, FSAN had consensus on the PON topology and the use of Time-Division Multiplexing/Multiple-Access (TDM/TDMA) for access control. However, there were many divergent areas of design. For this reason, standards offer several areas of implementation choice, such as the usage of single-fiber and two-fiber networks, the speed of transmission choices, and the optical budget classes. However, in 1996 the FSAN technical membership agreed that ATM (Asynchronous Transfer Mode) encapsulation of data would be the best solution for a multi-service network. This led to the first PON standard. This standard was adopted by ITU-T and was fully developed into a recommendation that eventually became ITU-T G.983.1, commonly referred to as ATM PON (A-PON).

Japan led the world in experimenting PONs and FTTx solutions. Nippon Telegraph and Telephone (NTT) deployed A-PON systems as a country-wide solution for broadband data access services. The system used by NTT was a single fiber (both directions of transmission on a single fiber strand), symmetric 155 Mb/s, class B system (25 dB of optical loss budget), and provided the user with an ATM-50 interface (an early interface standard that provided for 50 Mb/s of ATM access).

In the US, during the same time frame, several service providers had small-scale experimentations with PON systems on a trial or even a limited-service basis. BellSouth (now part of at&t) ran an early PON trial in the Dunwoody suburb of Atlanta, Georgia [8]. This trial was limited to a data-only overlay on top of the standard copper telephony network. The PON used in this trial was similar to that used in Japan, with two big exceptions. First, a two-level splitter arrangement was used in the distribution plant. Second, the ONUs for Dunwoody were built with Ethernet interfaces, rather than an ATM interface. This helped to avoid an expensive ATM-to-Ethernet converter device. The main objectives of this trial were to assess the maturity of the systems, and to get experience in deployment of PONs in the real-world network. The trial was a success in this limited sense by confirming that PON equipment was indeed field-ready, and that PON systems were tough enough to withstand the hardships of deployment. Yet, from the beginning it was obvious that PON was considerably more expensive than the other options available. As such, the trial did not progress to any wider deployments by BellSouth.

Meanwhile, Verizon and SBC (Southwestern Bell Communications, which is now also part of at&t) trialed A-PON systems, but for business applications and copper remediation uses. At that time, these carriers concluded that A-PON did not have significant cost advantages against new access-oriented SONET (Synchronous Optical Network) based platforms.

At the close of the 1990s, then, we could see that PON was a mature, deployable technology. However, the business case for PON deployment was lacking from several perspectives, especially in North America and Europe. First, the capital costs were still too high. Second, the demand for bandwidth in the amount that required fiber was far from obvious. Third, the competitive marketplace remained in a state of uncertainty. Fourth and lastly, the regulatory approach to new infrastructure had not been adjudicated by governments. Meanwhile, Cable MSOs were not yet into the triple-play mode of service delivery and competition. As a result of all these factors, most Telcos decided to focus on broadband growth scenarios that revolved around xDSL, leveraging the existing copper network. While such DSL networks were more limited in bandwidth (with most services of that era being below 1 Mb/s), they were inexpensive to deploy to the mass market, particularly in the incremental cost sense, and were much closer to the telephone operators experience base.

There were a few exceptions to this basic DSL trend. BellSouth constructed a fiber to the curb (FTTC) system to serve new-build areas. Despite the fact that it was a

proprietary system, it enjoyed some success, and more than one million customers are served by it. US West (now part of Qwest) also rolled out an A-PON FTTN system in Phoenix, Arizona, and Denver, Colorado. This early attempt to converge video, voice, and data services was a key stepping stone on the path to integrated networks and triple-play service capability. This system serves about 60,000 homes currently [9].

3. From A-PON to B-PON: Period of scaled deployment

In the year 2000, the telecommunications industry was heading to a period of steep downturn [10,11]. Indeed, it can be argued that the industry has not fully recovered yet from that downturn. Despite that situation, the Internet never ceased to grow throughout the first decade of the twenty first century. While this growth has not had any profound impact on the bandwidth-glutted carrier core networks, it did generate activities in the access domain and created more business interest in PON based solutions.

The A-PON system had been proven, but the most interested service providers (such as NTT and BellSouth) knew that it needed upgrades [12]. This led to A-PON evolving to Broadband PON (B-PON). B-PON was prompted by recommendations of FSAN which were later adopted by ITU as additions to the ITU-T G.983 series. These standards specified 622 Mb/s downstream, and 155 Mb/s or 622 Mb/s aggregate upstream data rates. Each OLT is shared by up to 32 ONUs for a maximum separation of 20 km between the OLT and ONU. B-PONs use TDM for multiple access and ATM cells for data framing. Therefore, some professionals continue to refer to B-PON as A-PON.

The G.983.3 standard specifies wavelength division duplex on a single fiber with the 1.3 μm wavelength window for upstream transmission and the 1.49 μm window for downstream. The 1.55 μm wavelength window is reserved for analog TV overlay. Another B-PON related standard is the J.186, which describes the transmitter, Erbium-Doped Fiber Amplifier (EDFA), and ONT components.

BellSouth wanted a residential system to provide broadcast video services without requiring a switched digital video system. Therefore, B-PON was appropriate. For competitive reasons, on the other hand, NTT needed a way to offer 100 Mb/s service over B-PON. Given that the upstream PON speed was 155 Mb/s, dynamic bandwidth allocation was required to allow users to burst to 100 Mb/s as needed. This work moved through FSAN and ITU, becoming the G.983.4 standard. In a similar time period, optics technology advanced to where the prevalent PON speed was 622 Mb/s downstream, 155 Mb/s upstream. With improvements like these, B-PON systems attained their final configuration.

In parallel, Verizon started to look at B-PON for its broader residential market where operational savings could be realized. The B-PON system was the focus of the 2003 Tri-Company (BellSouth, SBC and Verizon) PON technical specification, and has been deployed in volume in North America until 2007.

4. Evolution towards E-PON and G-PON

In March 2001, the IEEE 802.3 started the 802.3ah Ethernet in the First Mile (EFM) project. One of the charters of the project was to standardize the transport of Ethernet frames on PONs. In contrast to the approach of FSAN/ITU, the IEEE prefers a technology-driven development philosophy. The result was a simple system that began with existing Ethernet devices, and applied them to PONs with as little change as possible. The outcome was E-PON, which was ideal for carrying best-effort Internet access services. The IEEE 802.3ah E-PON standard was ratified in June 2004. It specifies a symmetric upstream/downstream line rate of 1 Gb/s and a transmission distance of 10 or 20 km with 16 ONUs per OLT. E-PON serendipitously came to market just as NTT had an urgent need for it. NTT needed a more advanced broadband solution to answer very strong competition from aggressive and financially backed ISP, and E-PON was the solution. Therefore, E-PON has gained a lot of popularity and success in East Asia, but not in the US [1,13].

In the same time E-PON was developed, year 2001, FSAN/ITU (ITU Study Group 15, SG15) was also working in a next-generation PON. The FSAN membership came to the conclusion that a system with greater efficiencies and more alignment with network developments was desirable. The effort in this regards focused on PON to carry packetized traffic more natively. Three possible systems were considered. The first was a sort of ATM system. The second was an Ethernet-framing system. The third was a physical layer framing system, which borrowed basic ideas from SDH/SONET and the Generic Framing Procedure (GFP), and incorporated different features to make sure that all services could be supported. It was this third approach, which is now referred to as the Gigabit-capable PON (G-PON) that was selected in the end and became the basis for the G.984 standards. G-PON uses the Generic Encapsulation Method or the G-PON Encapsulation Mode (GEM) and has increased the transmission rate to 2.5 Gb/s downstream and 1.25 Gb/s upstream.

By 2002, the major work of defining the G-PON standards was complete. However, the commercial production of the equipment would take several years. This considerable lag was caused by two factors. First, the optical and electronic devices needed to make G-PON work at the speeds desired (2.4 Gb/s downstream, 1.2 Gb/s upstream) took time to develop. Second, the maturity of the full-service schemes that ran over Ethernet and IP (Internet Protocol), not ATM or TDM, was not sufficient in 2002. G-PON ultimately offered a more capable system, and so it took longer to produce. US Telcos, such as Verizon and at&t, will rely on G-PON to compete with Cable MSOs in delivering triple play. G-PON became an attractive solution in Europe too, with deployments in England, France, Germany, Italy, and Spain, to name a few.

Thus, while the alignment of a simple need with a simple technology paved the way for mass deployment of E-PON in Japan, a strong case was made for the more sophisticated G-PON in the rest of the world where full service deemed desirable. This directed the choice of a PON

technology to be either E-PON or G-PON on geographical basis. Nevertheless, contrary to what names may suggest, support for Ethernet traffic is critical in all sorts of PONs. As a matter of fact, more than 90% of today's IP packets are generated and terminated as Ethernet frames. This includes data, voice, and video due to success of Voice over IP and migration of video streaming services to IP-based digital platforms. New video servers stream both broadcast and on-demand TV signals as Ethernet packets. Therefore, efficient handling of Ethernet packets is very important in all PON systems [1].

5. Current deployed systems

Hence, this history has left us in a situation where the world has two major PON standard systems: E-PON, which is based on Ethernet frame transmission and is described in the IEEE 802.3ah standard, and G-PON, which is based on GEM fragment transmission and is described in the ITU G.984 series. These two systems are intrinsically incompatible, and the market has evolved to deploy the two systems on a geographically diverse basis.

A comparison of these two architectures is given in Fig. 2, drawn in protocol stack style. On the left is the IEEE 802.3ah system, which defines the physical (PHY) and medium access control (MAC) layers in terms of 3 sub-layers each. On the right is the ITU-T G-PON system, which defines the PHY and gigabit transmission convergence (GTC) layers. Generally, in each sub-layer, there is a major design selection that needed to be made, and many of these are called out in the diagram. In nearly each of these decisions, the two standards groups took contrary positions. On the IEEE side, several of the sub-layers were inherited from prior point to point work. The remaining functions then were tailored to fit. Also at play was the strict scoping of the 802.3 process, which prevented the consideration of several functions, such as security or dynamic bandwidth allocation. On the ITU side, the design of the system was open to new ideas and could cover any subject matter that was seen as relevant to PONs. Nevertheless, many of the B-PON design points were inherited for the simple reason that they worked, and the designers were most familiar with them.

E-PON systems have primarily been used for data-only service offerings, and have received the most deployment in Japan and Korea. Telephony, broadcast video, or IPTV are provided via external boxes and networks, and the systems tend to be more aligned with the “over the top” service model. In a sense, the E-PON optical network unit (ONU) is a replacement for a DSL or cable modem. This eliminates a great deal of copper wiring issues, but it also raises the potential trouble of placing fiber cabling inside customer's home.

G-PON systems have been deployed mainly for full-service offerings in North America (e.g., Verizon's FiOS service and at&t's Lightspeed service), Europe (British Telecom, Deutsche Telekom, France Telecom, Telefonica, and Telecom Italia all have G-PON based deployed systems), and the Middle-East (Etisalat's). Video is being provided using a mixture of broadcast overlay and Internet protocol television (IPTV), typically in a hybrid

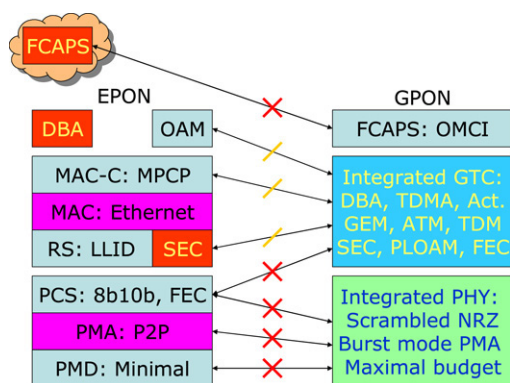


Fig. 2. A comparison of IEEE and ITU Gigabit rate PON standards.

Act.: Activation

DBA: dynamic bandwidth allocation

FCAPS: fault configuration accounting performance-monitoring security

GTC: G-PON transmission convergence

LLID: logical link identifier

MAC-C: MAC control

MPCP: Multi Point Control Protocol

OAM: operation and maintenance

OMCI: ONT management configuration interface

PCS: physical coding sublayer

PLOAM: physical layer OAM

PMA: physical medium attachment

PMD: physical media dependent

RS: réconciliation

SEC: security

arrangement. Essentially, the ONU becomes the new network interface device, and is a significant piece of equipment. Also, there is a trend to incorporate more and more functions into this equipment, which may become the customer's complete home networking device in future.

6. Next generation technologies

While most service providers believe that G-PON and E-PON will be more than sufficient to meet the anticipated demands and requirements for broadband access, they keep on the tradition of looking into their crystal balls and trying to envision the systems that will come next. This is driven by the desire to make sure that the system being deployed today will be expandable in future. In particular, the fiber infrastructure is very costly to build, and this must be reused in an economical upgrade. Therefore, both FSAN/ITU and IEEE have begun to consider systems that operate at aggregate speeds of 10 Gb/s, and that can do so using the existing PON infrastructure.

The IEEE 802.3av task force was established in 2007 to develop 10 Gb/s E-PON systems. That group has progressed work on two systems. The first is a 10 Gb/s downstream/1 Gb/s upstream. The second is a symmetric 10 Gb/s system. The 10/1 system is primarily seen as a transitional measure, to be used in certain near-term situations, and the symmetric system is the real objective that most practitioners want. However, 10 Gb/s upstream transmission appears to be technically and economically challenging, and it may be several years before it becomes practical.

The FSAN group has been considering many different upgrade systems. For instance, physically reduced split-ratio G-PONs were considered. In this model, capacity is increased by reducing the PON split ratio. Downstream-only WDM reduced split ratio G-PON (also known as "Stacked G-PON") was also considered. Both approaches were dropped however since they are both G-PON derivatives and they require PON reconfiguration. The following list enumerates other systems which are being pursued by FSAN, and gives a terse description of each system's disposition and the reasoning behind it (Note: XG = 10 Gb/s):

- (1) Bidirectional WDM reduced split ratio G-PON (also known as "Overlaid G-PON"). This reduces the split ratio in both directions using WDM. This remains a possibility; however, it is viewed as an interim step, and one that may quickly be overcome by the 10-G technologies.
- (2) XG-PON1 (10 Gb/s downstream, ~2.5 Gb/s upstream). This is viewed as a speed-enhanced version of G-PON, with various improvements to reflect the higher speed operation. FSAN has recently selected this system as their primary focus.
- (3) XG-PON2 (10 Gb/s symmetric). This symmetric system would likely be the ultimate version of TDMA PON. Recent moves in ITU and IEEE have indicated that this system would be based upon the 802.3av PHY and MAC layers, with ITU and Broadband Forum describing the higher layers.
- (4) Extended reach/split versions of the XG-PONs. A common theme to both XG-PON is to allow WDM-aware OLT and ONT components to permit multiplexing on the feeder fibers and also better application of optical amplifiers. This is being explored as an option to the base systems described in items 2 and 3.

The current round of PON standardization will not be finished until 2010 at the earliest, so the final outcome is still speculative. However, there have been several very productive liaison statement exchanges between ITU-T Study Group 15 and IEEE 802.3 regarding the harmonization of the two PON standards for the next generation of PON. If this trend continues, then we expect to see an arrangement of standards where compatibility issues are addressed. It is not possible to accurately envision how this would materialize. However, Fig. 3 depicts one possible scenario and many others are possible. There are two major systems in this case: XG-PON1 and XG-PON2. The asymmetric XG-PON1 is described primarily by ITU recommendations. The physical layer uses scrambled None-Return to Zero (NRZ). This should ensure the most economical optics possible. The TC (Transmission Convergence) layer is a modified G-PON TC. This would include mandatory Forward Error Correction (FEC) for improved link budget, and also a word-aligned framing concept that makes ASIC design simpler. There would also be further enhancements and streamlining of the ancillary protocols to adapt the system to the higher speed.

The symmetric XG-PON2 is described primarily by the IEEE 802.3av standards. The physical layer uses 64b66b scrambled block coding in both directions, and mandatory

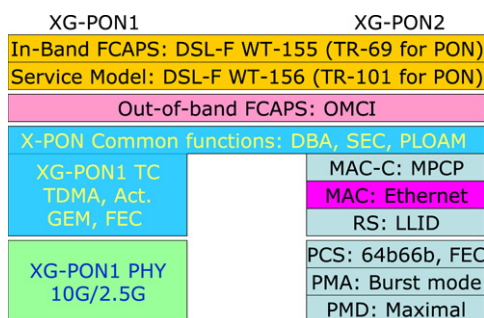


Fig. 3. The harmonized ITU/IEEE next-generation PON standards system.

strong FEC is used in both directions to meet the link budget requirements at 10 Gb/s operation. The TC layer is a modified multi-point control protocol (MPCP) coming from 1 Gb/s E-PON. The modifications here are quite minor, since the MPCP is more loosely coupled to the PHY layer than G-PON TC is.

Both of the systems share a set of standards that describe functions which are generic to both systems. There will be ITU recommendations for common TC related functions, such as dynamic bandwidth control and physical layer Operation and Maintenance (OAM). Also, out-of-band fault, configuration, accounting, performance monitoring, and security (FCAPS) management can be described in a new version of the ONT management and configuration interface (OMCI). Higher layer aspects, such as the service model and in-band management may be described in DSL-forum documents.

Finally, it should be pointed out that our discussion of next-generation technologies thus far does not include the sort of systems that are referred to in literature as WDM PONs. These systems (of which there are more than 10 distinct technology types) have been the focus of a huge amount of academic and industrial research, and hundreds of papers. To put these systems into perspective, it must be noted that they target bandwidths on the order of 40 Gb/s per PON. Therefore, these systems could be thought of as the “generation after next”. The tantalizing prospect of leap-frogging the 10 Gb/s generation has been entertained in many forums; however, the practical challenges for WDM PON systems are substantial. At the present time, such systems carry significant cost premiums. There has been progress on this front, but not enough to make it economical. Also, the current mass-market demand for bandwidth is not large enough to justify the capacity of WDM PONs. Therefore, most practitioners consider these PONs to be for future systems, but not for the ones that will be deployed soon.

7. What triggers the future? ...A US perspective

Before service providers spend the billions it takes to deploy new access systems, they need a business driver. The usual aphorisms do indeed apply here, as current and future deployments are usually motivated by hope, fear, and greed. Hope that the government may mandate deployment as a strategy to modernize the

country's network and drive economic growth. Fear that the competitors may threaten to take the existing service provider's customer base away, and something must be done to stop this threat. Greed, in that a service provider believes there is easy money to be made with a certain killer combination of services and networks.

Of course, feeding into these base motivations are the overall economy of telecommunications. There is the basic demand for raw bandwidth, for integrated services, and value-added special features. There is the competition between service providers for the different segments of telecommunications' users (residential, small/medium business, and large business). Lastly, there are organic and revolutionary technology advances that can change the cost and performance parameters of the business.

Considering the demand for bandwidth, everybody agrees that Internet traffic is growing. However, access revenues are flat at best. The biggest question is what is the new bandwidth for? So far, bandwidth has little intrinsic value in the traditional sense. Rather, customers just want to buy a fast pipe, and they seem to buy the fastest pipe possible, provided it does not cost more than they are willing to spend. They do this even though they do not really have a use for the bandwidth. Also, there is a range of customers for whom a lower-price/lower-performance product can sell alongside a premium product. With this background, it appears that bandwidth seems predominantly a competitive tool to obtain market share. The network that can provide the faster access at the same price wins. The slower network provider ends up having to cut price to stay in the game. So, nobody dies, service providers just have different profitability levels. Total service capability also factors into the equation. For example, if one competitor can provide triple-play services, then they all need this capability. Of course, such integration can be tackled via business cooperative agreements (e.g., Telco-Satellite television).

Advances in technology can also trigger new deployments. This is typically used by vendors to disrupt existing supplier relationships, and make space and opportunities for new equipment and systems. It is interesting to note that many times, the new technology was not strongly wanted, or at least not wanted by the incumbent service providers and suppliers. Still, new vendors of all categories try to advance their cause with new technologies. In the PON space currently, both 10 Gb/s PON and WDM PON are examples of technology that is intended to be disruptive.

8. Conclusion

Passive Optical Networks (PONs) have been the focus of considerable research, development, and standardization efforts over recent years. Today, they are well positioned as the technology of choice for broadband access in the near future. ATM based PONs (A-PON) was the first generation of PONs to be deployed and they evolved to Broadband PONs (B-PON). Today, Ethernet PONs (E-PON) and Gigabit-capable PON (G-PON) are the focus of service providers worldwide. Standardization organizations

are even working on next-generation systems and it is anticipated that the next round would be completed in the 2010 time frame. While it can be argued that E-PON and G-PON are sufficient for today's most mainstream applications, it is likely that competitive service provider(s) and/or aggressive equipment vendor(s) will break into markets with next generation PON systems in the foreseeable future.

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